DO NOW — RECALL

Compilers, interpreters and assemblers

Name: _____ Class: _____ Date: _____

1. What does LMC LDA do?

2. What is an executable?

3. What is a mnemonic?

4. Sorted [2,5,8,11,14,17,20,23,26], target23. Which midpoint indexes?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Loads value into ACC from mailbox.

2. Model answer

Machine-executable program representation.

3. Model answer

Human-readable instruction abbreviation.

4. Model answer

mid4→14, mid6→20, mid7→23; indexes 4,6,7.

Vocabulary — know these terms

compiler — Translates high-level source into target code.

interpreter — Runs a program through an execution runtime.

assembler — Translates assembly to machine encoding.

object code — Translator-generated machine-level representation.

NEW LEARNING

Read, annotate and explain

Understand

A compiler translates source code into another representation, often creating executable machine code via linking.

Apply

An interpreter executes program statements with a runtime, often without a separate standalone compiled executable.

Evaluate

An assembler translates assembly mnemonics and operands into processor machine instructions; translation strategies can overlap in modern environments.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A developer wants to distribute a compiled native executable. Which translator and what does it produce?

Predict first

A compiler translates the original language into target/object code

Model answer

linking resolves required components.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Identify compiler, interpreter and assembler by typical role. [3]

Question 2

Explain two differences between compiling native code and interpreting source at runtime. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate using an interpreter in a classroom programming exercise versus distributing a native compiled application. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An assembler translates assembly-language mnemonics into machine instructions.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.